AcadLink AI

Thematic Area of the Proposal

Academic Career Opportunity System with AI Integration

Abstract

This project proposes a centralized, AI-driven platform that bridges academic and career opportunities for students, faculty, and administrators. It aims to replace fragmented, outdated systems with a tailored, intelligent solution that enhances academic planning, job discovery, and institutional efficiency.

Current Status (National & International)

- **National Level:**
  - Heavily dependent on offline bulletin boards or institution-specific portals
  - Lack of unified academic-career planning tools
  - Minimal integration of AI in student support systems
- **International Level:**
  - Platforms like LinkedIn and ResearchGate serve professionals, not students specifically

Objectives

- Create a centralized digital ecosystem for academic and career opportunities
- Provide AI-powered recommendations tailored to academic profiles
- Facilitate role-based dashboards for students, faculty, and administrators
- Enable real-time tracking, notifications, and analytics

Identified Gaps

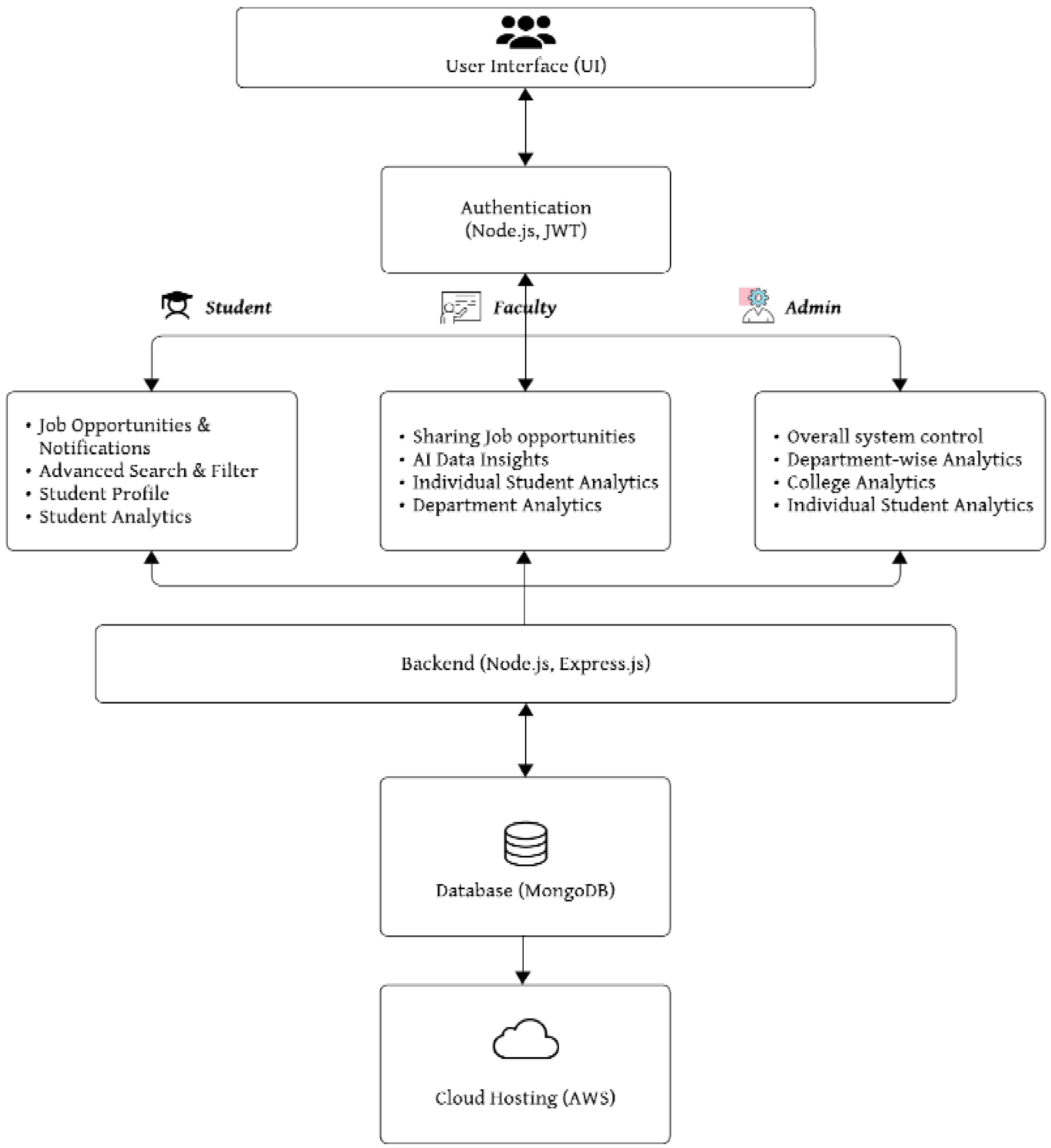
- No dedicated academic career platforms with localized features
- Lack of role-specific access and dashboards in existing systems
- Insufficient use of AI to support decision-making in academic paths

Novelty of the Work

- AI-powered academic guidance and opportunity recommendation engine
- Role-based personalized dashboards with actionable analytics
- Integrated platform combining internships, research, scholarships, and placements
- Faculty and admin tools to monitor and contribute to institutional growth

Architecture

- **Frontend**
  - React, Tailwind CSS, Next.js for responsive UI
- **Backend**
  - Node.js + Express API for business logic
- **Database**
  - MongoDB for user, opportunity, and analytics storage
- **Hosting**
  - AWS for scalability and reliability
- **Access Layers**
  - Student Module
  - Faculty Module
  - Admin Module
- **Security**
  - JWT-based authentication with role-specific access



Work in Phases & Expected Outcome

Work in Phases:

- Phase 1: Requirement Analysis & UI/UX Prototyping
- Phase 2: Core Development (Modules + AI Engine)
- Phase 3: Testing the developed modules
- Phase 4: Deployment & Feedback Integration

Expected Outcome:

- A live, scalable academic-career platform with real-time features
- Improved academic and placement outcomes
- Institutional analytics for continuous improvement

Timeline

Milestone	Timeline
Comprehensive Requirements Elicitation	Month 1
System Architecture Design	Month 2
Development Implementation	Months 3-9
Rigorous Testing and Deployment	Months 10-12

Budget

Item	Cost (₹ Lakhs)	Detailed Justification
Frontend Design & Development	0.20	Covers UI/UX design, development, and asset procurement. Figma Pro will be used for design, ensuring a professional and collaborative design experience at a reduced student cost.
Backend Engineering	0.30	Ensures a secure, scalable, and efficient system for user authentication, data storage, and API development.
Database Implementation & Hosting , Testing & Documentation	0.51	Covers MongoDB setup, configuration, and AWS hosting for high performance and security. Includes bug testing, security checks, and creation of user manuals and technical documentation.
AI Model & Infrastructure	0.30	OpenAI API, DeepSeek R1 (if fine-tuning), or custom model training with cloud GPUs (AWS, GCP, or local GPU setup).
Contingency Provisions	0.15	Allocated for unexpected challenges, technical issues, or necessary adjustments during development.
Total Estimated Cost	**₹1.46 Lakhs**	

Deliverables

- Fully functional AI-powered academic career platform
- Admin, Faculty, and Student dashboards
- Analytics and reporting features
- Training manuals and documentation
- Deployment and support infrastructure

References

- Exploring the impact of training and placement cell on career transition to employment. https://www.sciencedirect.com/science/article/pii/S219985312400177X

- A framework to improve university–industry collaboration. https://www.emerald.com/insight/content/doi/10.1108/jiuc-09-2019-0016/full/html

THANK YOU